



Instituto Federal de Santa Catarina
Campus Florianópolis

Introdução a Machine Learning

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Inteligência Artificial

- Sistemas Especialistas
- Machine Learning
- Algoritmo Genético
- Lógica Fuzzi
- Visão Computacional
- Raciocínio Baseado em casos
- Redes Neurais (Deep learning)



Machine Learning

O aprendizado de máquina (*Machine Learning* – ML) é um campo de estudo em ascensão, cujo principal objetivo é o desenvolvimento de algoritmos que permitem aos computadores aprender





Machine Learning

Ainda em expansão, o aprendizado de máquina é usado para ensinar computadores a executar uma ampla variedade de tarefas úteis. Isso inclui tarefas como a **detecção automática de objetos em imagens**, **reconhecimento de fala**, **descoberta de conhecimento nas ciências médicas** e **análises preditivas**.



Machine Learning

- Detecção de Fraudes
- Sistemas de Recomendação
- Mecanismos de Busca
- Sistemas de Vigilância em Vídeo
- Reconhecimento de Manuscrito
- Processamento de Linguagem Natural
- Bots de Serviço ao Cliente
- Manutenção Preditiva
- Detecção de Anomalia
- Previsão de Demanda
- Logística
- Diagnóstico de Saúde



Machine Learning

Considerado uma subárea da Inteligência Artificial (IA), o aprendizado de máquina utiliza o **raciocínio indutivo**, metodologia que extrai regras e padrões de grandes conjuntos de dados a fim de alcançar os resultados.

Outros métodos da IA também utilizam o raciocínio dedutivo, onde o conhecimento é baseado na lógica de regras pré-definidas.

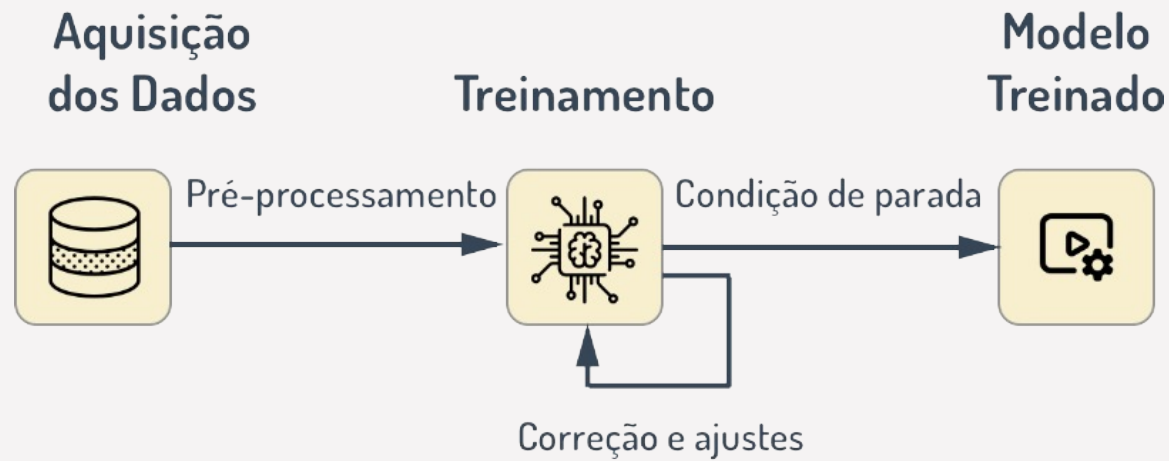


Machine Learning

- **Inteligência Artificial (IA)**
 - Abordagem Probabilística
 - Raciocínio Dedutivo
 - Raciocínio Indutivo
 - **Machine Learning (ML)**
 - Aprendizado Não Supervisionado
 - **Aprendizado Supervisionado**



Aprendizado Supervisionado





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Ferramentas





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Introdução a Machine Learning

Python

<https://www.python.org/>

The screenshot shows the Python.org homepage with a dark blue header and navigation bar. The header includes links for Python, PSF, Docs, PyPI, Jobs, and Community. Below the header is a search bar with a 'GO' button and a 'Socialize' link. The main content area features a 'Quick & Easy to Learn' section with a code snippet demonstrating Python 3 syntax for simple output and input/assignment. The code snippet is as follows:

```
# Python 3: Simple output (with Unicode)
>>> print("Hello, I'm Python!")
Hello, I'm Python!

# Input, assignment
>>> name = input('What is your name?\n')
>>> print('Hi, %s.' % name)
What is your name?
Python
Hi, Python.
```

Below the code snippet is a 'Quick & Easy to Learn' section with the text: 'Experienced programmers in any other language can pick up Python very quickly, and beginners find the clean syntax and indentation structure easy to learn. [Whet your appetite](#) with our Python 3 overview.' At the bottom of the page, there is a footer with the text: 'Python is a programming language that lets you work quickly and integrate systems more effectively. >>> [Learn More](#)'.



scikit-learning

<https://scikit-learn.org/stable/>

The screenshot shows the scikit-learn website homepage. At the top, there is a navigation bar with links for 'Install', 'User Guide', 'API', 'Examples', and 'More'. The main header features the 'scikit-learn' logo and the tagline 'Machine Learning in Python'. Below the header, there are three buttons: 'Getting Started', 'Release Highlights for 1.0', and 'GitHub'. To the right of these buttons, there is a list of bullet points describing the library's features: 'Simple and efficient tools for predictive data analysis', 'Accessible to everybody, and reusable in various contexts', 'Built on NumPy, SciPy, and matplotlib', and 'Open source, commercially usable - BSD license'. The main content area is divided into three columns: 'Classification', 'Regression', and 'Clustering'. Each column contains a brief description, applications, algorithms, and a representative figure. The 'Classification' column shows a grid of 12 small plots illustrating different classification models. The 'Regression' column shows a line plot titled 'Boosted Decision Tree Regression' comparing the performance of different models. The 'Clustering' column shows a scatter plot titled 'K-means clustering on the digits dataset (PCA-reduced data)' with centroids marked by white crosses.

scikit-learn
Machine Learning in Python

Getting Started Release Highlights for 1.0 GitHub

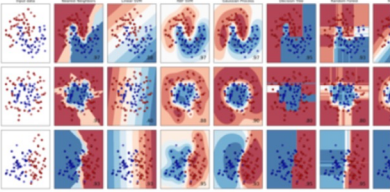
- Simple and efficient tools for predictive data analysis
- Accessible to everybody, and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib
- Open source, commercially usable - BSD license

Classification

Identifying which category an object belongs to.

Applications: Spam detection, image recognition.

Algorithms: SVM, nearest neighbors, random forest, and more...

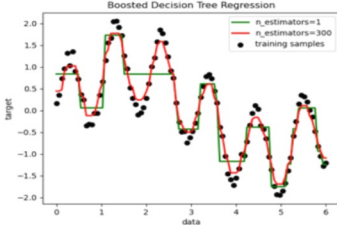


Regression

Predicting a continuous-valued attribute associated with an object.

Applications: Drug response, Stock prices.

Algorithms: SVR, nearest neighbors, random forest, and more...

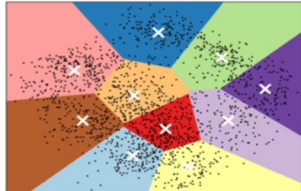


Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, Grouping experiment outcomes

Algorithms: k-Means, spectral clustering, mean-shift, and more...





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Introdução a Machine Learning

Jupyter Notebook

<https://jupyter.org/index.html>

The screenshot displays a web browser window with the Jupyter Slides interface. The browser's address bar shows the URL `localhost:8888/notebooks/Jupyter%20Slides.ipynb#`. The Jupyter interface includes a top menu bar with options like File, Edit, View, Insert, Cell, Kernel, Widgets, and Help. Below the menu is a toolbar with icons for various actions. The main content area is titled "Jupyter Notebook Slides Demonstration" and is attributed to "by Matt Speck". It contains an "Overview:" section with the text "Jupyter Notebooks can be easily converted into slideshows for presenting code." and "Notes for presentation". Below this is a code cell labeled "In [6]:" containing the following Python code:

```
import numpy as np
import matplotlib.pyplot as plt
X = np.random.uniform(0,50,100)
Y = np.random.uniform(0,30,100)

plt.plot(X,Y,'bo')
plt.show()
```

The code cell is followed by a scatter plot showing 100 data points (blue circles) distributed randomly within a 50x30 coordinate system. The x-axis ranges from 0 to 50, and the y-axis ranges from 0 to 30.



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Introdução a Machine Learning

Anaconda

<https://www.anaconda.com/products/individual>

The screenshot shows the Anaconda website's product page for the Individual Edition. At the top is a navigation bar with the Anaconda logo, links for Products, Pricing, Solutions, Resources, Blog, and Company, and a 'Get Started' button. The main content area features the Anaconda logo, the text 'Individual Edition', and the headline 'Your data science toolkit'. Below this is a paragraph describing the toolkit as an easy way to perform Python/R data science and machine learning on a single machine. To the right, there is a product card for 'Anaconda Individual Edition' with a 'Download' button for macOS, specifying Python 3.8, 64-bit, and a 440 MB graphical installer. At the bottom of the card, it says 'Get Additional Installers' and shows icons for Windows, macOS, and Linux.

ANACONDA Products Pricing Solutions Resources Blog Company [Get Started](#)

Individual Edition

Your data science toolkit

With over 25 million users worldwide, the open-source Individual Edition (Distribution) is the easiest way to perform Python/R data science and machine learning on a single machine. Developed for solo practitioners, it is the toolkit that equips you to work with thousands of open-source packages and libraries.

Anaconda Individual Edition

[Download](#) 

For MacOS
Python 3.8 • 64-Bit Graphical Installer • 440 MB

Get Additional Installers

 |  | 



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Introdução a Machine Learning

COLAB

<https://colab.research.google.com/>

The screenshot displays the Google Colaboratory (COLAB) web interface. At the top, the header reads "Olá, este é o Colaboratory" with navigation links for "Arquivo", "Editar", "Ver", "Inserir", "Ambiente de execução", "Ferramentas", and "Ajuda". On the right, there are links for "Compartilhar", "Conectar", "Editar", and a user profile icon.

The left sidebar contains an "Índice" (Index) with the following items:

- Primeiros passos
- Ciência de dados
- Machine learning
- Mais recursos
- Exemplos de machine learning
- Seção

The main content area is titled "Ciência de dados" (Data Science). It includes a text block stating: "Com o Colab, você pode aproveitar todo o potencial das conhecidas bibliotecas Python para analisar e ver dados. A célula de códigos abaixo usa **numpy** para gerar dados aleatórios e **matplotlib** para visualizá-los. Para editar o código, basta clicar na célula e começar a editar."

Below the text is a code cell containing the following Python code:

```
import numpy as np
from matplotlib import pyplot as plt

ys = 200 + np.random.randn(100)
x = [x for x in range(len(ys))]

plt.plot(x, ys, '-')
plt.fill_between(x, ys, 195, where=(ys > 195), facecolor='g', alpha=0.6)

plt.title("Sample Visualization")
plt.show()
```

The output of the code cell is a line plot titled "Sample Visualization". The x-axis represents 100 data points, and the y-axis ranges from 195 to 203. The plot shows a blue line representing the data points. A green shaded area is filled between the data points and the horizontal line at y=195, where the data points are above 195. The plot is titled "Sample Visualization".



Python



<https://www.python.org/>

- * Numbers
- * Strings
- * Boleanos
- * Printando

- * Listas
- * Dicionários
- * Tuplas
- * Sets

- * Operadores de comparação
- * if,elif, else
- * for Loops
- * while Loops
- * range()
- * list comprehension
- * funções
- * expressões lamda
- * map e filter
- * métodos



Python



<https://numpy.org/>



<https://pandas.pydata.org/>



<https://scipy.org/>



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Introdução a Machine Learning

Python

The screenshot shows a Jupyter Notebook interface with the following content:

```
In [3]: print("Olá mundo!")
Olá mundo!
```

```
In [5]: 1+1
Out[5]: 2
```

```
In [7]: a=2
        b=3
        c=a+b
        c
Out[7]: 5
```

```
In [8]: print(c)
5
```